



RTL Design Sherpa

RAPIDS Byte-Granular DMA Engine Byte Characterization Report

September 30, 2026

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Readme

Generated: 2026-10-02

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RAPIDS Byte-Granular DMA: Byte Characterization Report

Version: 0.10

Date: 2026-10-02

Platform: Genesys 2, 256-bit, 8 channels, 100 MHz

Results: rapids_byte_perf_20260930_142731.json

1. Headline

Largest transfers at 8 channels

Path	Point	Efficiency	MB/s measured	% of 3200 MB/s peak
Sink	byte path, 4096 B x 8 ch	1.000	357	11.1 %
Sink	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Sink	beat path, 4096 beats x 8 ch, word-wide checker build	1.000	3182	99.4 %
Source	byte path, 4096 B x 8 ch	1.000	352	11.0 %
Source	beat path, 4096 beats x 8 ch	1.000	356	11.1 %
Source	beat path, 4096 beats x 8 ch, word-wide checker build	1.000	3199	100.0 %

The byte-wise rates sit at the harness checker ceiling of $3200 / 9 = 355.6$ MB/s, not at the DUT's limit: the byte-wise CRC checkers take 9 cycles per 32-byte beat (section 3). Read every MB/s in this report against both the 3200 MB/s peak and that ceiling. Efficiency (payload over beats x lanes) does not depend on the checker.

The word-wide checker build (BYTE_CRC=0) takes one beat per cycle and is not checker-bound: its rows show the DUT's own rate against the 3200 MB/s peak. Its data check covers 4 of the 32 bytes of each beat, so it is a rate measurement and the byte-wise build is the integrity measurement.

2. Definitions

Definitions

Term	Meaning
Bytes moved	Total payload bytes over all active channels and descriptors; from the exact byte count of the AXIS bus meter.
Beats moved	Accepted beats on the interface named: AXIS beats on the stream side, memory beats on the AXI side.
Efficiency	$\text{payload bytes} / (\text{beats} \times \text{BYTE_LANES})$, $\text{BYTE_LANES} = 32$. 1.0 means every lane of every beat carried payload.
MB/s	Total bytes / measurement window, windows counted in 100 MHz cycles by the harness meters, not wall clock.
Peak	$3200 \text{ MB/s} = 100 \text{ MHz} \times 32 \text{ B}$, per direction. Shown beside every measured MB/s value.
Engaged utilization	$\text{prod} / (\text{prod} + \text{bp} + \text{starv})$ on one interface; the RAPIDS Beats headline metric, used only for the beat-aligned comparison.

3. Beat-aligned rows against the RAPIDS Beats report

These rows use the beat-scaled path (`pkt_bytes=None`, beats per channel) on the byte DUT, which is how the RAPIDS Beats report measured its matrix: 9-beat bursts, response delay 0, backpressure off, default seed, bare bus meters. The metric is engaged utilization ($\text{prod} / (\text{prod} + \text{bp} + \text{starv})$), the RAPIDS Beats headline metric. “Unchanged” is checked numerically: every cell is compared to the same cell of `genesys_dw256_obs_C.json` and must agree within 0.5 percentage points, the threshold the RAPIDS Beats report itself applies between builds.

3.1 Word-wide checker build (the verdict)

Measured on the `BYTE_CRC=0` bitstream (`BUILD.WORD_CRC = 1`, results file `rapids_byte_perf_20261002_032941.json`, bitstream sha256 `fdab4c4df63c02f7`). Its checkers take one beat per cycle like the RAPIDS Beats build. They CRC one 32-bit slice of each

beat (slice 0, which holds the beat's LFSR word) and ignore the strobes, so the golden is the RAPIDS Beats golden and the data check covers 4 of the 32 bytes of each beat. That is enough for a utilization measurement of whole-beat rows; full-byte integrity is what the byte-wise build in 3.2 and the rest of this report check. The design under test is the same RTL.

Each meter window opens on its own interface's first handshake (rd, wr, sout), and the sink-ingress window opens on the first ACCEPTED beat; cycles where the stream is offered before that accept are counted by OBS_SIN_LAUNCH (CSR 0x158) and are reported with the mechanism table below. The v0.2 report's rows were measured with every window on the shared obs_dut_busy open, which charged the descriptor-fetch wait to starvation on the memory-side interfaces; those start-up readings are harness artifacts and were corrected by the window change, not by a DUT change.

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	50.0 / 14.3	11.1 / 6.7	50.0 / 6.7	43.33
1	4	100.0 / 80.0	80.0 / 40.0	33.3 / 22.2	80.0 / 22.2	57.78
1	16	25.0 / 94.1	94.1 / 72.7	66.7 / 53.3	94.1 / 53.3	69.12
1	64	57.1 / 98.5	98.5 / 91.4	88.9 / 82.1	98.5 / 82.1	41.32
1	256	84.2 / 99.6	99.6 / 97.7	97.0 / 94.8	99.6 / 94.8	15.40
1	1024	95.5 / 99.9	99.9 / 99.4	99.2 / 98.7	99.9 / 98.7	4.38
1	4096	98.8 / 100.0	100.0 / 99.9	99.8 / 99.7	100.0 / 99.7	1.13
2	1	66.7 / 66.7	50.0 / 22.2	18.2 / 11.8	50.0 / 11.8	38.24
2	4	10.5 / 88.9	88.9 / 57.1	50.0 / 36.4	88.9 / 36.4	78.36
2	16	32.0 / 97.0	97.0 / 84.2	80.0 / 69.6	97.0 / 69.6	64.97
2	64	65.3 / 99.2	99.2 / 95.5	94.1 / 90.1	99.2 / 90.1	33.92
2	256	88.3 / 99.8	99.8 / 98.8	98.5 / 97.3	99.8 / 97.3	11.53
2	1024	96.8 / 100.0	100.0 / 99.7	99.6 / 99.3	100.0 / 99.3	3.16
2	4096	99.2 / 100.0	100.0 / 99.9	99.9 / 99.8	100.0 / 99.8	0.81
4	1	100.0 / 80.0	50.0 / 30.8	26.7 / 19.0	50.0 / 19.0	30.95

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
4	4	12.9 / 94.1	94.1 / 72.7	66.7 / 53.3	94.1 / 53.3	81.21
4	16	37.2 / 98.5	98.5 / 91.4	88.9 / 82.1	98.5 / 82.1	61.25
4	64	70.3 / 99.6	99.6 / 97.3	97.0 / 94.8	99.6 / 94.8	29.28
4	256	90.5 / 99.7	99.9 / 99.4	99.2 / 98.7	99.9 / 98.7	9.25
4	1024	97.4 / 99.9	100.0 / 99.9	99.8 / 99.7	100.0 / 99.7	2.50
4	4096	99.3 / 100.0	100.0 / 100.0	100.0 / 99.9	100.0 / 99.9	0.64
8	1	4.0 / 88.9	50.0 / 38.1	34.8 / 27.6	50.0 / 27.6	84.85
8	4	14.5 / 97.0	97.0 / 84.2	80.0 / 69.6	97.0 / 69.6	82.42
8	16	40.5 / 99.2	99.2 / 95.5	94.1 / 90.1	99.2 / 90.1	58.72
8	64	73.1 / 99.8	99.8 / 98.5	98.5 / 97.3	99.8 / 97.3	26.66
8	256	91.6 / 96.1	100.0 / 99.7	99.6 / 99.3	100.0 / 99.3	4.51
8	1024	97.8 / 99.0	100.0 / 99.9	99.9 / 99.8	100.0 / 99.8	1.24
8	4096	99.4 / 99.7	100.0 / 100.0	100.0 / 100.0	100.0 / 100.0	0.32

Verdict: CHANGED. 112 of 112 beat-aligned cells were compared; 85 differ by more than 0.5 pp; the largest difference is -84.85 pp (ch 8, 1 beats, sink sin). Cells over the threshold: ch1 b1 sink wr +35.71 pp; ch1 b1 source rd +4.44 pp; ch1 b1 source sout +43.33 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr +40.00 pp; ch1 b4 source rd +11.11 pp; ch1 b4 source sout +57.78 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr +21.39 pp; ch1 b16 source rd +13.33 pp; ch1 b16 source sout +40.78 pp; ch1 b64 sink sin -41.32 pp;

Where the differences come from. Every difference is a fixed number of cycles per run, not a change of rate. The cycle counts below are the evidence; the same counts at 16 beats per channel are identical to these. Only the 1-beat rows differ: the write and source starvation there equals the channel count (channel count + 7 on the read side), a per-channel start-up term that later beats overlap, and the sink backpressure is 0 at 1 to 4 channels.

Cycle counts at 4096 beats per channel (fixed start-up terms, not rates)

Ch	Sink AXIS-in bp, byte	beats	Sink AXI4-wr starv, byte	beats	Source AXI4-rd starv, byte	beats
1	47	0	1	6	8	14
2	67	0	1	6	8	14
4	107	2	1	6	8	14
8	187	82	1	6	8	14

- **Sink AXIS-in.** The byte ingress accepts a channel's stream only once that channel has a packet record, because the destination offset and the expected length come from the descriptor (sink ingress chapter of the MAS). While one channel occupies the stream the others are offered data with t ready low, and the meter counts those cycles as backpressure: exactly $27 + 20 \times \text{channels}$, the per-channel packet-record wait of rapids TASK-021, independent of the beats per channel. RAPIDS Beats has no such term because it buffers the stream before the descriptor arrives. The sin window opens at the first ACCEPTED beat. The wait this window choice could hide is measured and empty: OBS_SIN_LAUNCH (CSR 0x158) counts cycles where the stream is offered before the first accept, and it read 0 on all 28 points of this run (and on all 28 rows of the word-wide sim campaign), so opening at the first offered beat and at the first accepted beat are the same measurement on this DUT.
- **Sink AXI4-wr.** The write window opens at the first write handshake, so the wait above is not in it. Start-up starvation is 1 cycle at 16 beats and up, the same for every channel count (at 1 beat it is the channel count). Under the previous shared window the same cells read 9 to 17 cycles: the descriptor-fetch wait was being charged to write starvation, and the per-interface windows removed that, not the DUT.
- **Source.** The read and stream-out windows open on their own first handshakes: 8 starvation cycles on rd and 1 on sout at 16 beats and up, every channel count (channel count + 7 and channel count at 1 beat). These cells read 15 to 22 cycles under the shared window, for the same reason as the write side. The remaining small POSITIVE deltas against the beats reference at short transfers are the same class of fixed start-up term on the beats harness's side of the comparison.

- **Amortisation.** Because each term is fixed, the byte and beats readings converge as the transfer grows: at 4096 beats per channel every cell is within 1.13 percentage points, and the 8-channel row within 0.32. The 1-to-64-beat rows, where a fixed term is a large share of the window, are the ones that move most.
- **Not explained.** With 1 beat per channel and 1, 2 or 4 channels the byte build shows no sink AXIS-in backpressure, while the same channels at 16 beats and up do, and the 8-channel 1-beat row reads 190 cycles against the 187 of the formula. Not isolated.

3.2 Byte-wise checker build (the standard bitstream)

Engaged utilization, byte build / RAPIDS Beats reference, beat-aligned rows

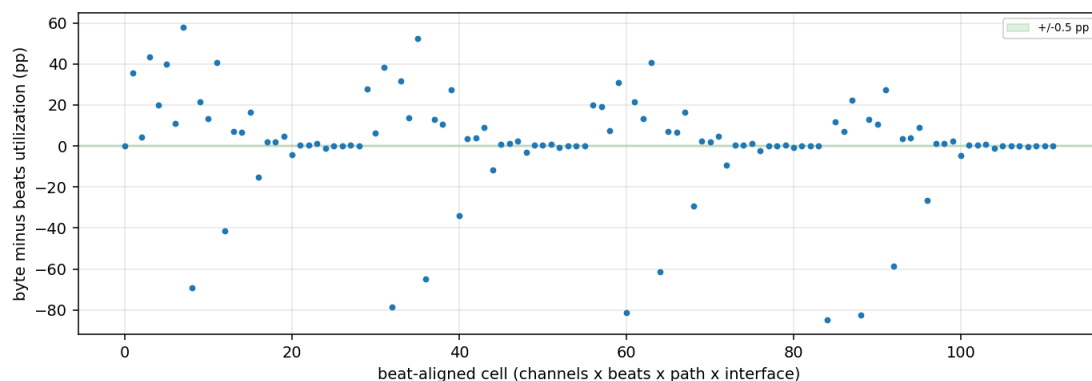
Ch	Beats/ch	Sink AXIS-in % (byte / beats)	Sink AXI4-wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
1	1	50.0 / 50.0	10.0 / 14.3	6.2 / 6.7	6.7 / 6.7	4.29
1	4	100.0 / 80.0	25.0 / 40.0	9.3 / 22.2	9.5 / 22.2	20.00
1	16	25.0 / 94.1	13.6 / 72.7	10.6 / 53.3	10.7 / 53.3	69.12
1	64	57.1 / 98.5	11.6 / 91.4	20.7 / 82.1	10.7 / 82.1	79.79
1	256	22.6 / 99.6	11.2 / 97.7	18.6 / 94.8	11.0 / 94.8	86.47
1	1024	12.7 / 99.9	11.1 / 99.4	12.4 / 98.7	11.1 / 98.7	88.27
1	4096	11.5 / 100.0	11.1 / 99.9	11.4 / 99.7	11.1 / 99.7	88.73
2	1	66.7 / 66.7	16.7 / 22.2	8.0 / 11.8	8.7 / 11.8	5.56
2	4	10.5 / 88.9	19.5 / 57.1	10.1 / 36.4	10.3 / 36.4	78.36
2	16	32.0 / 97.0	12.2 / 84.2	11.7 / 69.6	10.6 / 69.6	72.00
2	64	65.3 / 99.2	11.4 / 95.5	33.6 / 90.1	10.8 / 90.1	84.15
2	256	14.8 / 99.8	11.2 / 98.8	19.8 / 97.3	11.1 / 97.3	87.67
2	1024	11.8 / 100.0	11.1 / 99.7	12.5 / 99.3	11.1 / 99.3	88.58
2	4096	11.3 / 100.0	11.1 / 99.9	11.4 / 99.8	11.1 / 99.8	88.81
4	1	100.0 /	25.0 / 30.8	9.3 / 19.0	9.8 / 19.0	20.00

Ch	Beats/ch	Sink AXIS- in % (byte / beats)	Sink AXI4- wr %	Source AXI4-rd %	Source AXIS-out %	Max abs delta (pp)
		80.0				
4	4	12.9 / 94.1	14.2 / 72.7	10.6 / 53.3	10.7 / 53.3	81.21
4	16	37.2 / 98.5	11.6 / 91.4	21.0 / 82.1	10.8 / 82.1	79.79
4	64	70.3 / 99.6	11.2 / 97.3	50.3 / 94.8	10.8 / 94.8	86.12
4	256	12.6 / 99.7	11.1 / 99.4	20.7 / 98.7	11.1 / 98.7	88.27
4	1024	11.5 / 99.9	11.1 / 99.9	12.6 / 99.7	11.1 / 99.7	88.73
4	4096	11.2 / 100.0	11.1 / 100.0	11.4 / 99.9	11.1 / 99.9	88.85
8	1	4.0 / 88.9	16.0 / 38.1	10.1 / 27.6	10.4 / 27.6	84.85
8	4	14.5 / 97.0	12.5 / 84.2	11.3 / 69.6	10.6 / 69.6	82.42
8	16	40.5 / 99.2	11.4 / 95.5	34.5 / 90.1	10.8 / 90.1	84.15
8	64	73.1 / 99.8	11.2 / 98.5	66.2 / 97.3	11.0 / 97.3	87.30
8	256	11.8 / 96.1	11.1 / 99.7	21.4 / 99.3	11.1 / 99.3	88.58
8	1024	11.3 / 99.0	11.1 / 99.9	12.6 / 99.8	11.1 / 99.8	88.81
8	4096	11.2 / 99.7	11.1 / 100.0	11.5 / 100.0	11.1 / 100.0	88.87

Result on this build: CHANGED. 112 of 112 beat-aligned cells were compared; 108 differ by more than 0.5 pp; the largest difference is -88.87 pp (ch 8, 4096 beats, sink wr). Cells over the threshold: ch1 b1 sink wr -4.29 pp; ch1 b4 sink sin +20.00 pp; ch1 b4 sink wr -15.00 pp; ch1 b4 source rd -12.92 pp; ch1 b4 source sout -12.70 pp; ch1 b16 sink sin -69.12 pp; ch1 b16 sink wr -59.17 pp; ch1 b16 source rd -42.74 pp; ch1 b16 source sout -42.67 pp; ch1 b64 sink sin -41.32 pp; ch1 b64 sink wr -79.79 pp; ch1 b64 source rd -61.34 pp;

What the byte-checker rows can and cannot show. The byte build's on-chip checkers (BYTE_CRC=1 on axi4_slave_wr_crc_check and axis4_slave_pattern_check) fold the strobed bytes of each beat into the CRC four bytes per cycle and hold ready low while they do, so a 32-byte beat occupies 9 cycles. That caps the harness at $3200 / 9 = 355.6$ MB/s (11.1 % of the 3200 MB/s peak), which is where the large-transfer rows sit. The RAPIDS Beats reference used the word-wide checkers and is not checker-bound. The differences in this table are therefore the checkers backpressuring the DUT, not evidence about the DUT's own throughput, and this table is **not comparable** to the RAPIDS Beats report. The "utilization unchanged" criterion is settled in section 3.1 on the word-wide checker build (BYTE_CRC=0, BUILD.WORD_CRC = 1), where every checker takes one beat per cycle as in the RAPIDS Beats build.

Figure 3.1: Byte minus beats utilization, beat-aligned cells



utilization delta

Beat-aligned rows: bytes, beats, efficiency, MB/s and share of the 3200 MB/s peak

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
1	1	32	1	1.00 0	320	10.0 %	32	1	1.00 0	200	6.2 %
1	4	128	4	1.00 0	800	25.0 %	128	4	1.00 0	298	9.3 %
1	16	512	16	1.00 0	430	13.4 %	512	16	1.00 0	339	10.6 %
1	64	2048	64	1.00 0	372	11.6 %	1056	33	1.00 0	342	10.7 %
1	256	8192	256	1.00 0	359	11.2 %	4864	152	1.00 0	353	11.0 %
1	1024	32768	1024	1.00 0	357	11.1 %	29440	920	1.00 0	355	11.1 %
1	4096	131072	4096	1.00 0	356	11.1 %	127744	3992	1.00 0	355	11.1 %
2	1	64	2	1.00 0	533	16.7 %	64	2	1.00 0	256	8.0 %
2	4	256	8	1.00 0	337	10.5 %	256	8	1.00 0	324	10.1 %
2	16	1024	32	1.00	389	12.2	928	29	1.00	340	10.6

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink % peak	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
				0		%			0		%
2	64	4096	128	1.00	363	11.4	1312	41	1.00	344	10.8
				0		%			0		%
2	256	1638	512	1.00	357	11.2	9152	286	1.00	354	11.1
		4		0		%			0		%
2	1024	6553	2048	1.00	356	11.1	5830	1822	1.00	355	11.1
		6		0		%	4		0		%
2	4096	2621	8192	1.00	356	11.1	2549	7966	1.00	356	11.1
		44		0		%	12		0		%
4	1	128	4	1.00	800	25.0	128	4	1.00	298	9.3
				0		%			0		%
4	4	512	16	1.00	413	12.9	512	16	1.00	339	10.6
				0		%			0		%
4	16	2048	64	1.00	372	11.6	1056	33	1.00	346	10.8
				0		%			0		%
4	64	8192	256	1.00	358	11.2	1760	55	1.00	346	10.8
				0		%			0		%
4	256	3276	1024	1.00	357	11.1	1753	548	1.00	355	11.1
		8		0		%	6		0		%
4	1024	1310	4096	1.00	356	11.1	1158	3620	1.00	355	11.1
		72		0		%	40		0		%
4	4096	5242	1638	1.00	356	11.1	5090	1590	1.00	356	11.1
		88	4	0		%	56	8	0		%
8	1	256	8	1.00	129	4.0	256	8	1.00	324	10.1
				0		%			0		%
8	4	1024	32	1.00	397	12.4	960	30	1.00	338	10.6
				0		%			0		%
8	16	4096	128	1.00	363	11.4	1280	40	1.00	345	10.8
				0		%			0		%
8	64	1638	512	1.00	357	11.1	2720	85	1.00	352	11.0
		4		0		%			0		%

Ch	Beats /ch	Sink bytes	Sink beats	Sink eff	Sink MB/s	Sink	Src bytes	Src beats	Src eff	Src MB/s	Src % peak
						% peak					
8	256	6553	2048	1.00	356	11.1	3392	1060	1.00	355	11.1
		6		0		%	0		0		%
8	1024	2621	8192	1.00	356	11.1	2305	7204	1.00	356	11.1
		44		0		%	28		0		%
8	4096	1048	3276	1.00	356	11.1	1016	3178	1.00	356	11.1
		576	8	0		%	960	0	0		%

Every beat-aligned row moves whole beats, so efficiency is 1.000 by construction; measured range over 56 rows: 1.000 to 1.000.

4. Transfer size (offset 0)

One descriptor per channel, payload in bytes, start offset 0. Bytes moved and beats are totals over the active channels; efficiency is payload bytes over beats times 32 bytes per beat, on the AXIS side and on the memory side. MB/s is total bytes over the measurement window (the longer of the stream and memory windows) and is always shown with its share of the 3200 MB/s peak (100 MHz x 32 B).

4.1 Sink path

Size sweep, sink path, offset 0

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	10.0	0.31 %	PASS
1	0	2	2	2	2	0.031	0.031	16.7	0.52 %	PASS
1	0	4	4	4	4	0.031	0.031	25.0	0.78 %	PASS
1	0	8	8	8	8	0.031	0.031	4.0	0.13 %	PASS
7	0	1	7	1	1	0.219	0.219	70.0	2.19 %	PASS
7	0	2	14	2	2	0.219	0.219	116.7	3.65 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
7	0	4	28	4	4	0.219	0.219	175.0	5.47 %	PASS
7	0	8	56	8	8	0.219	0.219	28.1	0.88 %	PASS
32	0	1	32	1	1	1.000	1.000	320.0	10.00 %	PASS
32	0	2	64	2	2	1.000	1.000	533.3	16.67 %	PASS
32	0	4	128	4	4	1.000	1.000	800.0	25.00 %	PASS
32	0	8	256	8	8	1.000	1.000	128.6	4.02 %	PASS
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	0	2	66	4	4	0.516	0.516	471.4	14.73 %	PASS
33	0	4	132	8	8	0.516	0.516	113.8	3.56 %	PASS
33	0	8	264	16	16	0.516	0.516	129.4	4.04 %	PASS
64	0	1	64	2	2	1.000	1.000	533.3	16.67 %	PASS
64	0	2	128	4	4	1.000	1.000	914.3	28.57 %	PASS
64	0	4	256	8	8	1.000	1.000	220.7	6.90 %	PASS
64	0	8	512	16	16	1.000	1.000	251.0	7.84 %	PASS
77	0	1	77	3	3	0.802	0.802	550.0	17.19 %	PASS
77	0	2	154	6	6	0.802	0.802	208.1	6.50 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
77	0	4	308	12	12	0.802	0.802	256.7	8.02 %	PASS
77	0	8	616	24	24	0.802	0.802	290.6	9.08 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	0	2	406	14	14	0.906	0.906	431.9	13.50 %	PASS
203	0	4	812	28	28	0.906	0.906	386.7	12.08 %	PASS
203	0	8	1624	56	56	0.906	0.906	367.4	11.48 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS
1024	0	2	2048	64	64	1.000	1.000	371.7	11.62 %	PASS
1024	0	4	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
1024	0	8	8192	256	256	1.000	1.000	359.5	11.23 %	PASS
4035	0	1	4035	127	127	0.993	0.993	360.9	11.28 %	PASS
4035	0	2	8070	254	254	0.993	0.993	358.0	11.19 %	PASS
4035	0	4	16140	508	508	0.993	0.993	356.1	11.13 %	PASS
4035	0	8	32280	1016	1016	0.993	0.993	355.6	11.11 %	PASS
4096	0	1	4096	128	128	1.000	1.000	363.4	11.36 %	PASS
4096	0	2	8192	256	256	1.000	1.000	359.5	11.23 %	PASS

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
409 6	0	4	1638 4	512	512	1.000	1.000	357.5	11.17 %	PASS
409 6	0	8	3276 8	1024	1024	1.000	1.000	356.5	11.14 %	PASS

4.2 Source path

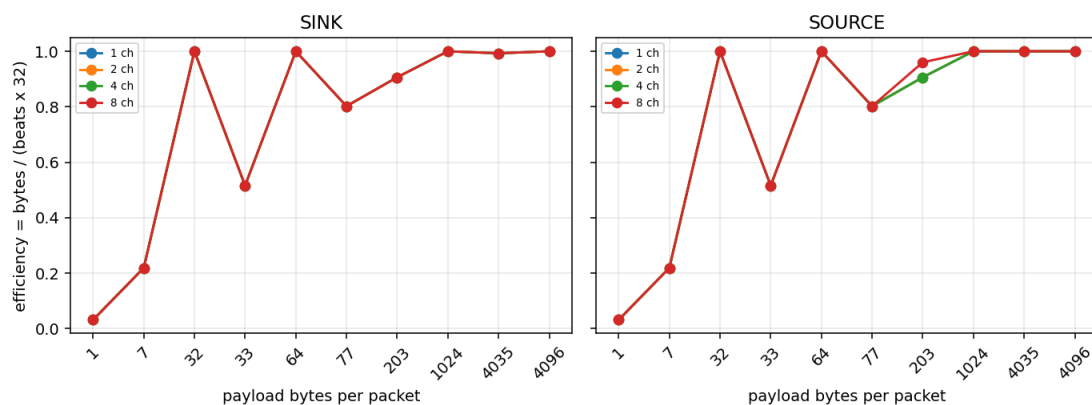
Size sweep, source path, offset 0

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1	0	1	1	1	1	0.031	0.031	6.2	0.20 %	PASS
1	0	2	2	2	2	0.031	0.031	11.1	0.35 %	PASS
1	0	4	4	4	4	0.031	0.031	18.2	0.57 %	PASS
1	0	8	8	8	8	0.031	0.031	26.7	0.83 %	PASS
7	0	1	7	1	1	0.219	0.219	43.8	1.37 %	PASS
7	0	2	14	2	2	0.219	0.219	73.7	2.30 %	PASS
7	0	4	28	4	4	0.219	0.219	112.0	3.50 %	PASS
7	0	8	56	8	8	0.219	0.219	151.4	4.73 %	PASS
32	0	1	32	1	1	1.000	1.000	200.0	6.25 %	PASS
32	0	2	64	2	2	1.000	1.000	256.0	8.00 %	PASS
32	0	4	128	4	4	1.000	1.000	297.7	9.30 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
32	0	8	256	8	8	1.000	1.000	324.1	10.13 %	PASS
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	0	2	66	4	4	0.516	0.516	183.3	5.73 %	PASS
33	0	4	132	8	8	0.516	0.516	227.6	7.11 %	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09 %	PASS
64	0	1	64	2	2	1.000	1.000	256.0	8.00 %	PASS
64	0	2	128	4	4	1.000	1.000	297.7	9.30 %	PASS
64	0	4	256	8	8	1.000	1.000	324.1	10.13 %	PASS
64	0	8	512	16	16	1.000	1.000	339.1	10.60 %	PASS
77	0	1	77	3	3	0.802	0.802	226.5	7.08 %	PASS
77	0	2	154	6	6	0.802	0.802	270.2	8.44 %	PASS
77	0	4	308	12	12	0.802	0.802	299.0	9.34 %	PASS
77	0	8	616	24	24	0.802	0.802	315.9	9.87 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	0	2	406	14	14	0.906	0.906	317.2	9.91 %	PASS
203	0	4	812	28	28	0.906	0.906	332.8	10.40 %	PASS

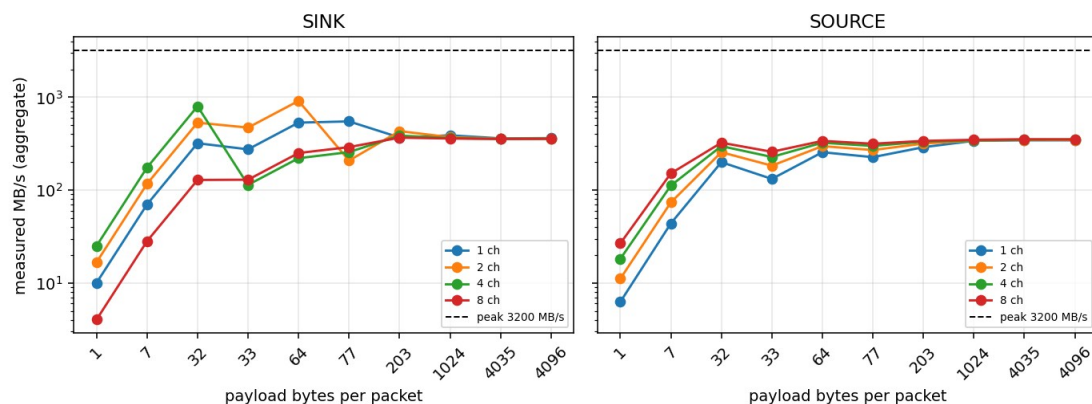
Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
203	0	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
1024	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS
1024	0	2	1056	33	64	1.000	0.516	341.7	10.68 %	PASS
1024	0	4	1280	40	128	1.000	0.312	343.2	10.72 %	PASS
1024	0	8	1760	55	256	1.000	0.215	348.5	10.89 %	PASS
4035	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	0	2	1760	55	254	1.000	0.217	347.1	10.85 %	PASS
4035	0	4	2656	83	508	1.000	0.163	349.0	10.91 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4096	0	1	1280	40	128	1.000	0.312	344.1	10.75 %	PASS
4096	0	2	1760	55	256	1.000	0.215	347.1	10.85 %	PASS
4096	0	4	2688	84	512	1.000	0.164	352.3	11.01 %	PASS
4096	0	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 4.1: Efficiency against payload size



efficiency vs size

Figure 4.2: Measured MB/s against payload size, peak as the dashed line



MB/s vs size

Observations from this data:

- Sink, 8 channels, 4096 B: 357 MB/s = 11.1 % of the 3200 MB/s peak at efficiency 1.000.
- Sink, 8 channels: 1 B payload reaches 4.0 MB/s, 89x less than 4096 B; the window is dominated by fixed per-packet latency (199 cycles for one beat per channel).
- Source, 8 channels, 4096 B: 352 MB/s = 11.0 % of the 3200 MB/s peak at efficiency 1.000.
- Source, 8 channels: 1 B payload reaches 26.7 MB/s, 13x less than 4096 B; the window is dominated by fixed per-packet latency (30 cycles for one beat per channel).

5. Start offset

A descriptor may start mid-beat. The offset moves payload into an extra memory beat when offset + payload crosses a beat boundary, lowering memory-side efficiency while the AXIS side, which packs from lane 0, is unchanged. Offset 0 rows repeat the size sweep.

5.1 Sink path

Offset sweep, sink path

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	1	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	5	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
33	31	1	33	2	2	0.516	0.516	275.0	8.59 %	PASS
203	0	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	1	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	5	1	203	7	7	0.906	0.906	369.1	11.53 %	PASS
203	31	1	203	7	8	0.906	0.793	225.6	7.05 %	PASS
1024	0	1	1024	32	32	1.000	1.000	389.4	12.17 %	PASS
1024	1	1	1024	32	33	1.000	0.970	317.0	9.91 %	PASS
1024	5	1	1024	32	33	1.000	0.970	318.0	9.94 %	PASS
1024	31	1	1024	32	33	1.000	0.970	324.1	10.13 %	PASS
403	0	1	4035	127	127	0.993	0.993	360.9	11.28	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
5									%	
4035	1	1	4035	127	127	0.993	0.993	360.9	11.28	PASS
5									%	
4035	5	1	4035	127	127	0.993	0.993	361.2	11.29	PASS
5									%	
4035	31	1	4035	127	128	0.993	0.985	344.6	10.77	PASS
5									%	
33	0	8	264	16	16	0.516	0.516	129.4	4.04	PASS
									%	
33	1	8	264	16	16	0.516	0.516	129.4	4.04	PASS
									%	
33	5	8	264	16	16	0.516	0.516	129.4	4.04	PASS
									%	
33	31	8	264	16	16	0.516	0.516	129.4	4.04	PASS
									%	
203	0	8	1624	56	56	0.906	0.906	367.4	11.48	PASS
									%	
203	1	8	1624	56	56	0.906	0.906	367.4	11.48	PASS
									%	
203	5	8	1624	56	56	0.906	0.906	368.3	11.51	PASS
									%	
203	31	8	1624	56	64	0.906	0.793	249.8	7.81	PASS
									%	
1024	0	8	8192	256	256	1.000	1.000	359.5	11.23	PASS
									%	
1024	1	8	8192	256	264	1.000	0.970	328.6	10.27	PASS
									%	
1024	5	8	8192	256	264	1.000	0.970	328.7	10.27	PASS
									%	
1024	31	8	8192	256	264	1.000	0.970	329.5	10.30	PASS
									%	

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
403 5	0	8	3228 0	1016	1016	0.993	0.993	355.6	11.11 %	PASS
403 5	1	8	3228 0	1016	1016	0.993	0.993	355.6	11.11 %	PASS
403 5	5	8	3228 0	1016	1016	0.993	0.993	355.7	11.12 %	PASS
403 5	31	8	3228 0	1016	1024	0.993	0.985	347.4	10.86 %	PASS

5.2 Source path

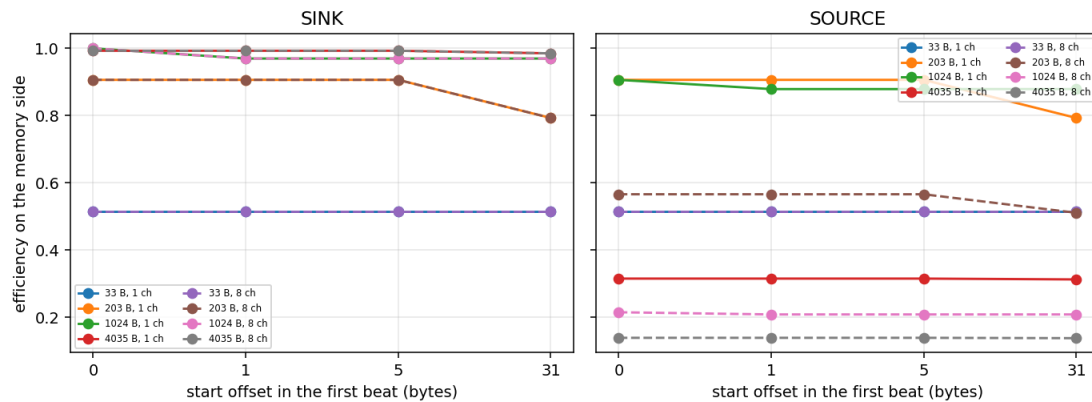
Offset sweep, source path

Payl oad B	Offset	Ch	Bytes move d	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
33	0	1	33	2	2	0.516	0.516	132.0	4.12 %	PASS
33	1	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	5	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
33	31	1	33	2	2	0.516	0.516	126.9	3.97 %	PASS
203	0	1	203	7	7	0.906	0.906	290.0	9.06 %	PASS
203	1	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	5	1	203	7	7	0.906	0.906	285.9	8.93 %	PASS
203	31	1	203	7	8	0.906	0.793	285.9	8.93 %	PASS
102 4	0	1	928	29	32	1.000	0.906	339.9	10.62 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	1	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
1024	5	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
1024	31	1	928	29	33	1.000	0.879	339.9	10.62 %	PASS
4035	0	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	1	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	5	1	1280	40	127	1.000	0.315	344.1	10.75 %	PASS
4035	31	1	1280	40	128	1.000	0.312	344.1	10.75 %	PASS
33	0	8	264	16	16	0.516	0.516	258.8	8.09 %	PASS
33	1	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
33	5	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
33	31	8	264	16	16	0.516	0.516	256.3	8.01 %	PASS
203	0	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	1	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	5	8	1014	33	56	0.960	0.566	339.1	10.60 %	PASS
203	31	8	1046	34	64	0.961	0.511	344.1	10.75 %	PASS
1024	0	8	1760	55	256	1.000	0.215	348.5	10.89 %	PASS

Payload B	Offset	Ch	Bytes moved	AXIS beats	Mem beats	Eff (AXIS)	Eff (mem)	MB/s	% of peak	Result
1024	1	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS
1024	5	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS
1024	31	8	1760	55	264	1.000	0.208	345.1	10.78 %	PASS
4035	0	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	1	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	5	8	4512	141	1016	1.000	0.139	353.3	11.04 %	PASS
4035	31	8	4512	141	1024	1.000	0.138	352.0	11.00 %	PASS

Figure 5.1: Memory-side efficiency against start offset



efficiency vs offset

Offsets that cost an extra memory beat (sink, 1 channel): 203 B at offset 31: 7 -> 8 memory beats; 1024 B at offset 1: 32 -> 33 memory beats; 1024 B at offset 5: 32 -> 33 memory beats; 1024 B at offset 31: 32 -> 33 memory beats; 4035 B at offset 31: 127 -> 128 memory beats.

6. Descriptor chains

Several descriptors per channel back to back (4 channels), offset 0. Efficiency counts payload over all beats of the chain.

Descriptor chains, 4 channels

Paylo ad B	Descs	Path	Bytes moved	AXIS beats	Eff (AXIS)	MB/s	% of peak	Result
1	16	sink	64	64	0.031	5.4	0.17 %	PASS
1	16	source	64	64	0.031	37.2	1.16 %	PASS
64	4	sink	1024	32	1.000	223.6	6.99 %	PASS
64	4	source	1024	32	1.000	347.1	10.85 %	PASS
64	16	sink	4096	128	1.000	221.4	6.92 %	PASS
64	16	source	3872	121	1.000	352.6	11.02 %	PASS
1024	4	sink	16384	512	1.000	341.4	10.67 %	PASS
1024	4	source	2656	83	1.000	350.9	10.96 %	PASS
1024	16	sink	65536	2048	1.000	336.3	10.51 %	PASS
1024	16	source	50432	1576	1.000	355.4	11.11 %	PASS

7. Source backpressure (integrity only)

The source backpressure is paced by the host toggling the checker's ready over UART, so the cycle counts measure the host, not the DUT (rapids TASK-085). These rows are integrity checks: bytes and beats must still match and the golden CRCs must agree. Their MB/s is not reported.

Source with backpressure, integrity rows

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
1	33	0	33	2	n/a (host- paced)	PASS
1	33	5	33	2	n/a (host-	PASS

Ch	Payload B	Offset	Expected bytes	Expected AXIS beats	MB/s	Result
					paced)	
1	1024	0	1024	32	n/a (host-paced)	PASS
1	1024	5	1024	32	n/a (host-paced)	PASS
4	33	0	132	8	n/a (host-paced)	PASS
4	33	5	132	8	n/a (host-paced)	PASS
4	1024	0	4096	128	n/a (host-paced)	PASS
4	1024	5	4096	128	n/a (host-paced)	PASS

8. Failures, caveats and coverage

No measured point failed.

Coverage: 117 of 117 planned points of profile standard are in this file (117 passed, 0 failed).

Standing limitations of the design, not of this measurement:

- The sink `s_axis_tready` is one signal qualified by TID, so a beat for a channel whose packet record has not arrived blocks every channel behind it on the stream (head-of-line blocking, inherent and documented).
- TYPE=EXT descriptors stay beat-aligned by design and are not part of the byte sweeps.
- AXI error responses are now exercised and the paths are proven on silicon (rapids TASK-020, 2026-10-01); this entry previously recorded them as unproven and the hook as unbuilt. See section 8.1.
- Each interface has its own measurement window, opened on that interface's first handshake; the `sin` window opens on the first ACCEPTED stream beat, and cycles where the stream is offered before that accept are counted by `OBS_SIN_LAUNCH` (section 3.1). MB/s here uses the longer of the stream and memory windows.

8.1 AXI response-error injection (rapids TASK-020)

The two synthetic AXI slaves are built `ERR_INJECT=1` in this harness, so a chosen burst on a chosen channel answers `SLVERR` instead of `OKAY`. `ERR_INJ (0x0C8)` arms it – `WR_EN` for the sink's B channel, `RD_EN` for the source's R channel, with `RESP`, `CH`, `SKIP` and `ONESHOT` – and `ERR_STAT (0x0CC)` reports from the SLAVE that the error was really issued, so a passing status check cannot be the DUT agreeing with itself. `ERR_INJECT=0` everywhere else keeps every other consumer answering `OKAY` as before.

Run	Result
Board, --byte-seq axi_resp_error --seq-level full, 8 channels	36/36 checks PASS , 7,969 UART ops
UART sim, same sequence at gate	18/18 checks PASS
Bitstream	sha256 754c3c2d . . . , WNS +0.401 ns at 100 MHz, 78,426 LUTs (38.5 %), 68 BRAM tiles

Checked per half, per round, with a second channel running alongside throughout: the slave issued the error (`ERR_STAT`); the sticky per-channel flag raised on the `TARGETED` channel only (`SNK_SCHERR` / `SRC_SCHERR` and `SCHED_ERROR`); `CHANNEL_RESET` cleared it; and both channels were golden against the byte-wise CRC afterwards. So the DUT's sticky `r_wr_error` (`BRESP`) and `r_rd_error` (`RRESP`) paths, and recovery through channel reset, are measured on hardware rather than inferred from unit simulation.

NOT covered, and deliberately not claimed: the monbus error `PACKET`. The host has no monbus-buffer readout – `MON_BASE/MON_LIMIT` are configured and never read back – so the packet half of the error contract needs a readout that does not exist yet. rapids TASK-020 keeps that box open.

8.2 Build variants measured against each other (2026-10-01)

One harness RTL; a variant is generic overrides, built by the named targets make `bitstream-{std,perf,mon,obs,ila}`. Every number below is parsed from that build's own post-route reports by `reports/extract_build_metrics.py` – none is typed in – and every variant was then programmed and run on the board.

Variant	LUT	vs std	BRAM	WNS ns	WHS ns	Failing eps	sha256
std (BYTE _CRC= 1)	78,426	–	68	0.401	0.045	0	754c3 c2dcd b7
perf	71,643	-6,783	52	0.426	0.037	0	1f268 d9896

Variant	LUT	vs std	BRAM	WNS ns	WHS ns	Failing eps	sha256
(BYTE_CRC=0)							9d
mon (USE_AXI_MONIT0 RS=1 GEN_MON=1)	84,190	+5,764	68	0.768	0.026	0	0678e8d52a88
obs (USE_OBSERVERS=1 + taps)	100,396	+21,970	68	0.215	0.031	0	4ff79d23b873

All four close timing with zero failing endpoints out of ~264 k.

What the comparison is for – the cost of instrumentation, which was previously unmeasured on the byte design:

- **The interface observers are expensive:** +21,970 LUTs, +28 % over std, and they more than halve the timing margin (0.401 -> 0.215 ns). That is the one variant where the margin is thin enough to matter on a tighter part.
- **The in-core monitors are cheaper than the observers by 4x** (+5,764 LUTs) and the build closed BETTER than std (0.768 ns). Place-and-route variation is part of that, so read it as “no margin cost”, not as an improvement.
- **The word-wide perf build is the cheapest in both LUTs and BRAM** (52 vs 68 tiles): the byte-wise CRC machinery is what costs the BRAM, which is the resource price of measuring integrity rather than rate.

Board results, same bitstreams:

Variant	Runs
std / mon / obs	byte smoke PASS, axi_resp_error 18/18 , byte-perf quick 7/7
perf	aligned profile 28/28 , 3182 MB/s sink /

Variant	Runs
	3199 MB/s source

The perf row reproduces section 1’s headline figures with a clean identity record on both ends of the run. An earlier attempt produced the same numbers but false-failed its END identity check – the Genesys 2 chain transiently lists the board with no device behind it – which marked good results “not from one board”; board_guard now re-reads the chain once before failing, and a test covers both that recovery and that a genuinely absent board still fails.

perf deliberately runs only the aligned profile: the host refuses byte-wise checks on a word-wide build (“cannot check a partial strobe”), which is correct and is why the integrity and rate measurements need two bitstreams.

9. Build and resources

Post-route resources and timing at 100 MHz, XC7K325T (203,800 LUTs, 445 BRAM tiles)

Build	Slice LUTs	Slice regs	BRAM tiles	WNS setup	WHS hold	Failing endpoint s	Bitstream sha256
bytec rc	92,874	82,379	68	+0.301 ns	+0.023 ns	0	cfad34c 3465f11 10
word crc	86,065	79,527	52	+0.077 ns	+0.013 ns	0	48e1282 c46e707 cd
perf	71,651	54,643	52	+0.431 ns	+0.049 ns	0	fdab4c4 df63c02 f7

Build configuration of each row:

- **bytecrc:** BYTE_CRC=1 USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=0 (channel-reset + hold-junk fixes).
- **wordcrc:** BYTE_CRC=0 USE_OBSERVERS=1 OBS_ENABLE_MON_TAPS=0. RTL commit d1d20b046.
- **perf:** BYTE_CRC=0 USE_AXI_MONITORS=0 GEN_MON=0 USE_OBSERVERS=0. RTL commit 8ae53d3ae.

Every build closes timing at 100 MHz with no failing endpoint. The slack is small and positive; treat 100 MHz as the design point, not as margin. The resource figures come from the post-route utilization report; the word-wide checker build is a measurement bitstream and is not the standard one.

10. Provenance and reproduction

Provenance

Item	Value
Results file	rapids_byte_perf_20260930_142731.json
Timestamp	2026-09-30T14:27:36
Profile	standard
Status	final
Bitstream	rapids_byte.bit sha256 cfad34c3465f1110
Design	256-bit, 8 channels, 4096 B SRAM per channel, 100 MHz, peak 3200 MB/s per direction
Beats reference	genesys_dw256_obs_C.json

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	cfad34c3465f1110	0x52415042	0x0A070820	0x00000000	0x00000000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.

One command runs the campaign and regenerates this report:

```
cd projects/fpga-systems/Genesys2/rapids/flows-rapids
./byte_perf.sh --profile full --final      # final numbers (standard
byte-CRC bitstream)
./byte_perf.sh --profile standard          # quick run, writes
*_prelim*.json
```

The word-wide aligned results are rapids_byte_perf_20261002_032941.json; its device readback follows.

Device readback per session

Session	Bitstream sha256	CSR_ID	BUILD	Sentinel start	Sentinel end	Stable	Aborted
1	fdab4c4 df63c02 f7	0x52415 042	0x18070 820	0x00000 000	0x00000 000	yes	no

The sentinel is the MON_LIMIT register written once at configure time. It resets on any FPGA reconfiguration, so a change between readbacks means the device was reprogrammed during the run. A point measured across such a change is dropped, not recorded. Device stable for the whole run: yes.